More Height-Balanced Trees

B-Trees and B^+-Trees

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Recap

In the previous classes, we discussed two height-balanced **binary** search trees, namely, the AVL tree and the red-black tree.

Today

Today we will discuss two height-balanced **m-ary** search trees, namely, the B-tree and the B^+-tree.

Development of the Height-Balanced Binary Search Trees: Recap

Year	Event
Late 1950s	**The binary search tree** was independently proposed by several researchers.
1962	**The AVL tree** was proposed by two Soviet mathematicians Adel'son-Vel'skiˇı (pronounced velskii) and Landis.
1972	**The red-black tree** was proposed by a German computer scientist Rudolf Bayer. At that time, Bayer called it the 'symmetric binary B-trees'.
1978	**The red and black colour convention** was introduced by two American computer scientists Guibas and Sedgewick.

Development of B-Trees

Year	Event
1972	Rudolf Bayer and Edward McCreight, while working at Boeing Research Labs, proposed **B-trees** for managing file indices. It was published in issue 3 of volume 1 of *Acta Informatica* [1].
1972	In issue 4 of volume 1 of *Acta Informatica*, Bayer proposed the 'symmetric **binary** B-trees' [2] that would later be known as **the red-black trees**.
1979	Douglas Comer (at Purdue University) published a survey of B-trees and its variants, including **B^+-trees**, in *ACM Computing Surveys*. Comer mentioned that B+-trees had been in use in IBM's virtual storage access method (VSAM). However, there is no single paper proposing the notion of B^+-trees.

[1] https://link.springer.com/article/10.1007/BF00288683
[2] https://link.springer.com/article/10.1007/BF00289509
[3] https://dl.acm.org/doi/pdf/10.1145/356770.356776

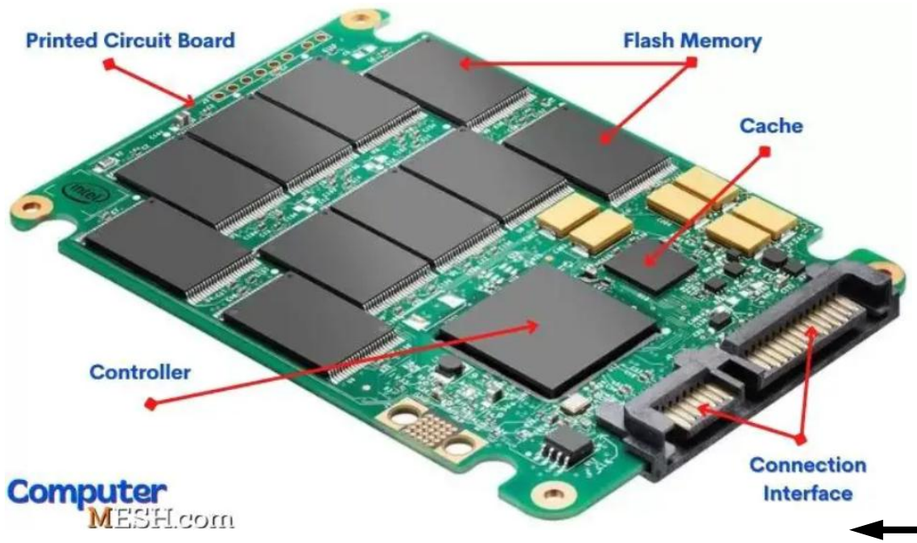
B-Trees

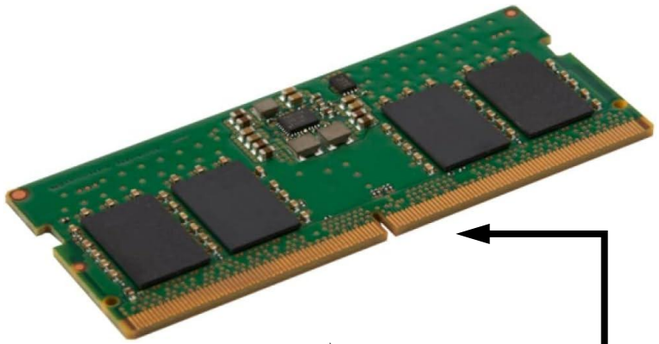
Nobody knows what the ‘B’ in B-tree stands for.
Bayer and McCreight kept it a mystery.

B-Trees

A B-tree is a height-balanced search tree designed to accommodate a **secondary storage**.

Typical Memory Hierarchy in a Computer





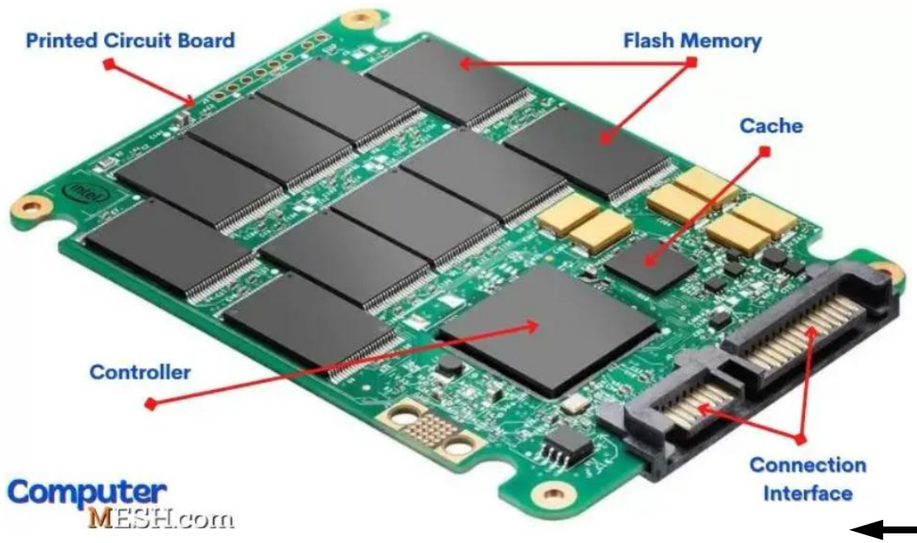
Secondary Storage (disk drive)
A typical NVMe PCIe SSD has a bandwidth of only **3.5 Gbps**. Substantially slower than the main memory.

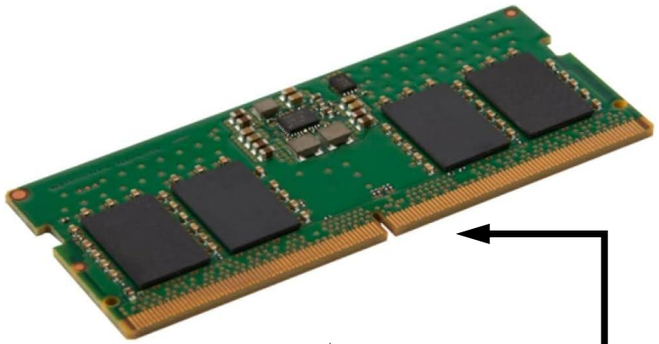
Main Memory (RAM)
A typical DDR5 SDRAM has a bandwidth of **51.2 Gbps**.

CPU

CPU: https://circuitdigest.com/news/dhruv64-debuts-as-india-first-indigenous-10-ghz-64-bit-dual-core-processor
RAM: https://www.amazon.in/HP-DDR5-SDRAM-Memory-Module/dp/B0BSLPTSKD

Typical Memory Hierarchy in a Computer (contd.)





A secondary storage typically has a capacity in TBs. It usually divides its content into 512B **blocks**.

When a block is requested by the CPU, the block is copied from the secondary storage to the main memory. Subsequently, CPU performs reads/writes on that block. If a write operation is performed on the block, the block is copied back to the secondary storage.

CPU

CPU: https://circuitdigest.com/news/dhruv64-debuts-as-india-first-indigenous-10-ghz-64-bit-dual-core-processor
RAM: https://www.amazon.in/HP-DDR5-SDRAM-Memory-Module/dp/B0BSLPTSKD

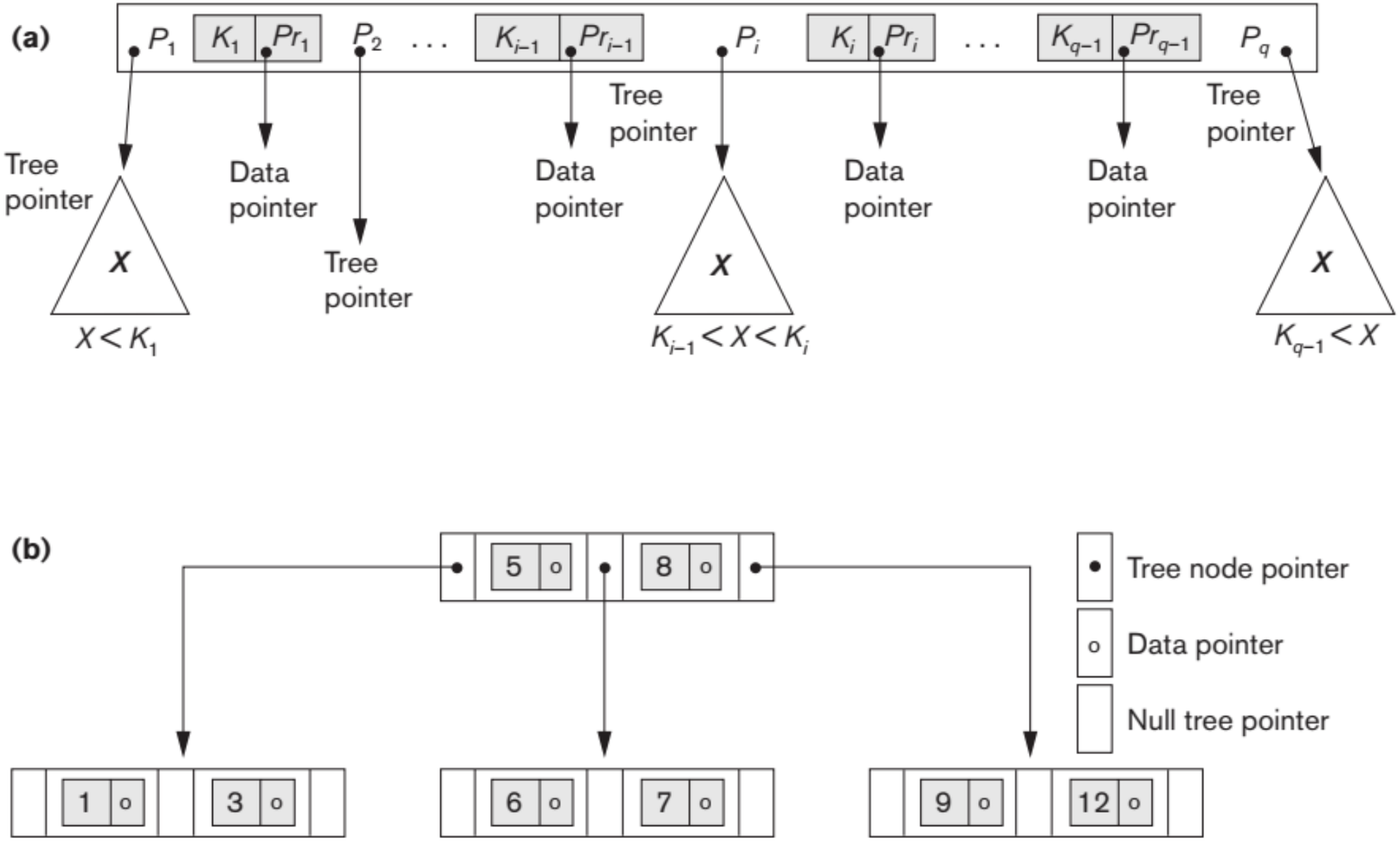
B-Trees (contd.)

Since the secondary storage is substantially slower than the main memory, we should **minimise the number of block transfers to and from the disk**.

If we only keep one key in a node, the search tree would become quite large and would require a large number of blocks.

Hence, a B-tree stores as many keys as possible in a node. Ideally, a node should be the size of a block. Suppose, a block is 512B and each key (along with its satellite data pointer) requires 8B. Then each node can accommodate (512 / 8) = 64 keys at maximum.

B-Trees (contd.)



Here, we assume that each node can have at most 2 keys. In practice, a B-tree node can have 2,000 keys.

Figure 17.10, Fundamentals of Database Systems, Elmasri and Navathe, 7th Edition, 2016

B^+-Trees

B^+-Trees

A B^+-tree is one of the natural variants of the B-trees. It has been widely used for organising file indices.

In a B^+-tree, the **satellite data pointers** are stored only in the **leaf nodes**.

Structures of the B$^+$-Trees (a) Internal and (b) Leaf Nodes

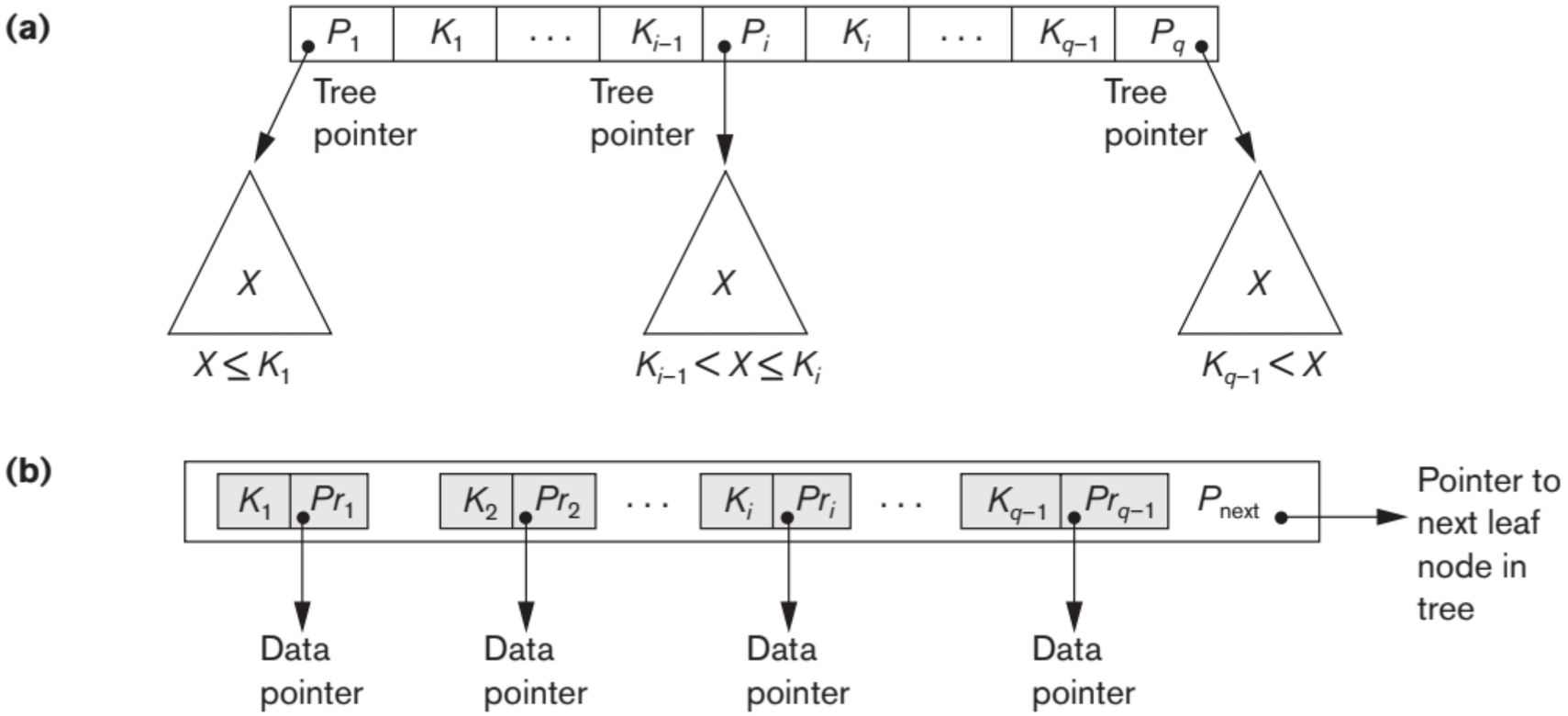


Figure 17.11, Fundamentals of Database Systems, Elmasri and Navathe, 7th Edition, 2016

B^+-Trees: An Example

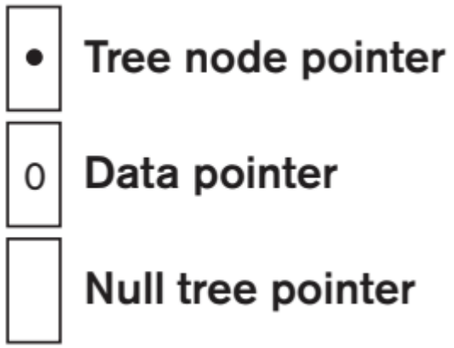
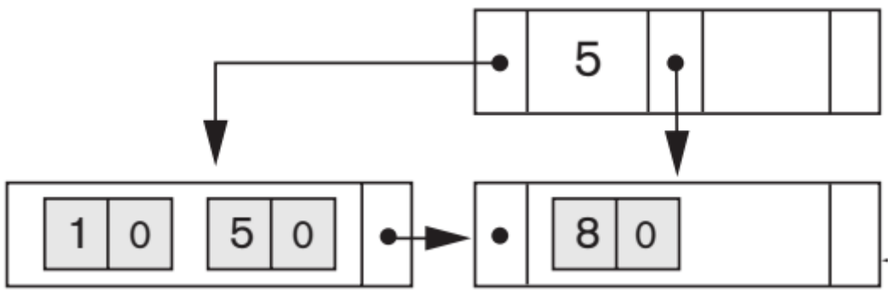


Figure 17.12, Fundamentals of Database Systems, Elmasri and Navathe, 7th Edition, 2016

Insert 1 into the B$^+$-Tree: An Overflow Occurrs



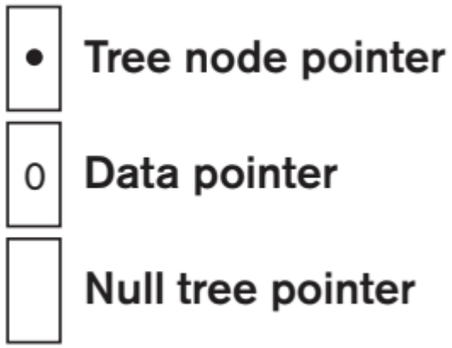
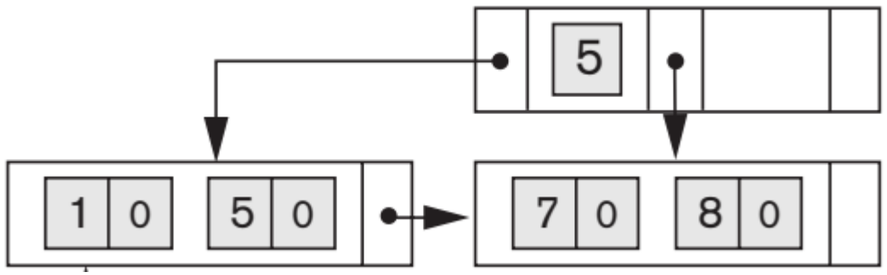


Figure 17.12, Fundamentals of Database Systems, Elmasri and Navathe, 7th Edition, 2016

Insert 1 into the B$^+$-Tree: Insert 7



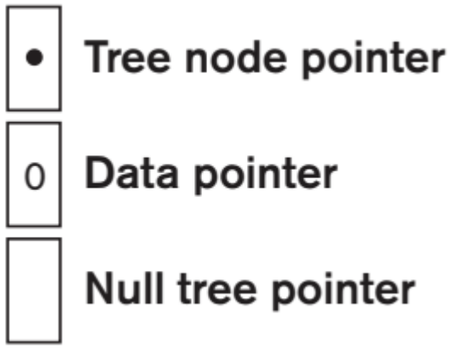


Figure 17.12, Fundamentals of Database Systems, Elmasri and Navathe, 7th Edition, 2016

Insert 1 into the B$^+$-Tree: Insert 3, Overflow

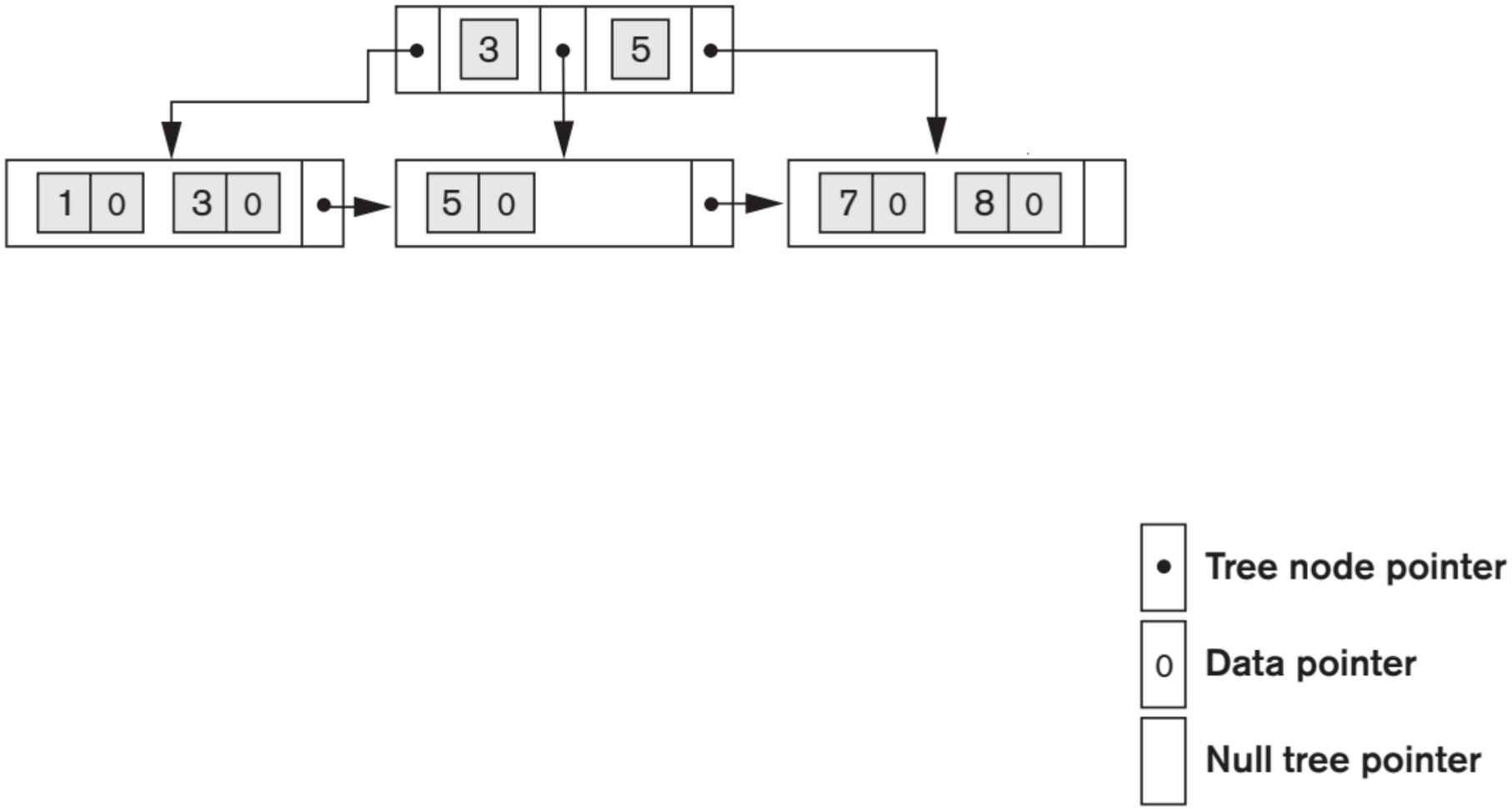


Figure 17.12, Fundamentals of Database Systems, Elmasri and Navathe, 7th Edition, 2016

Insert 1 into the B$^+$-Tree: Insert 12, Overflow

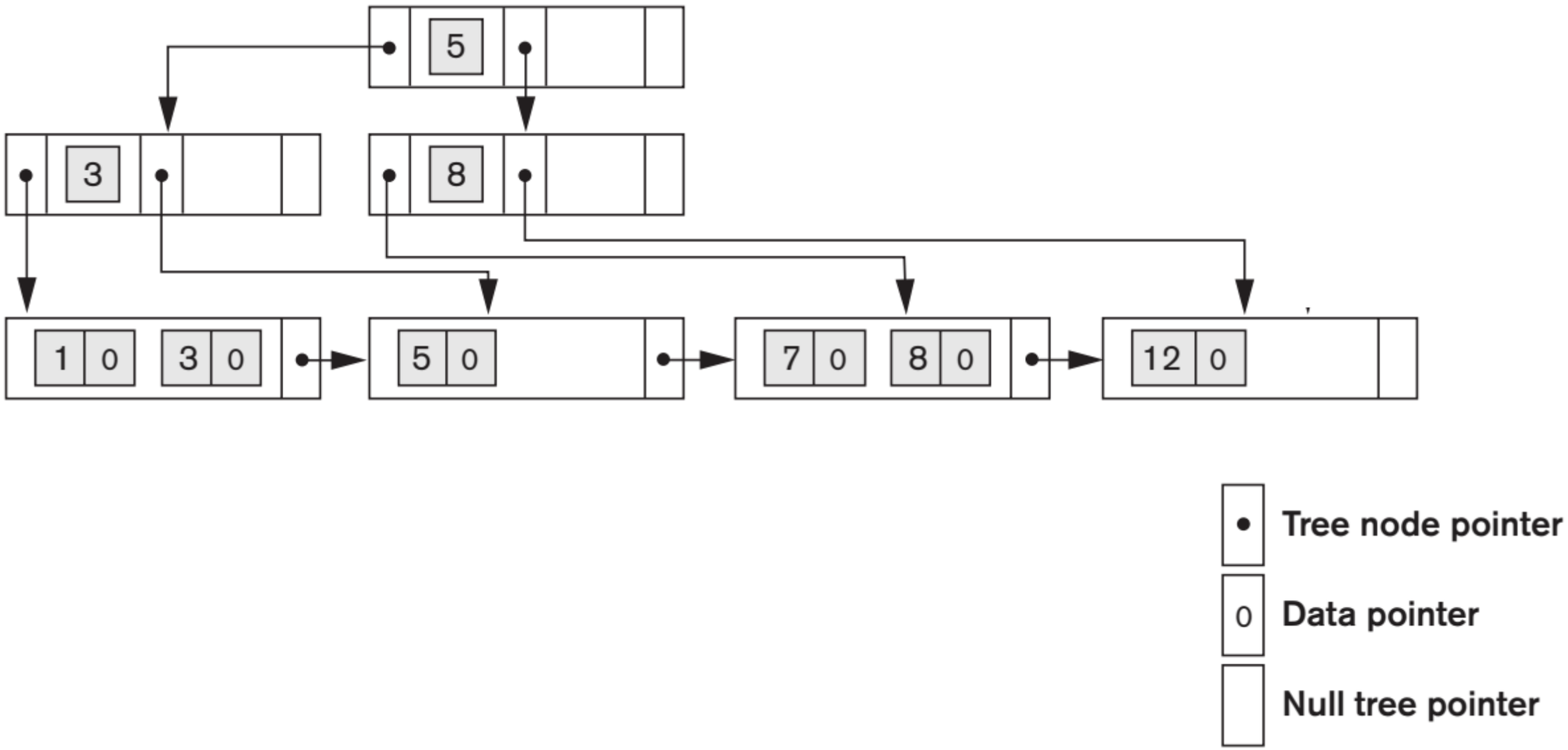
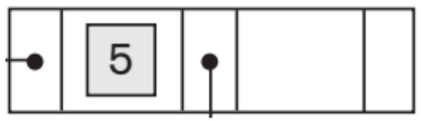
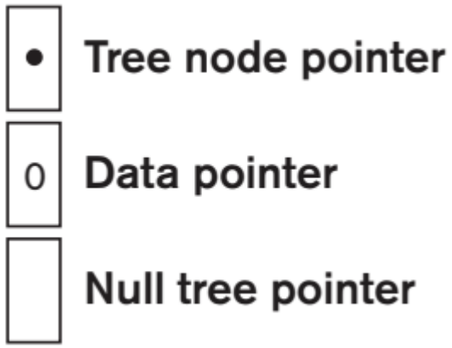


Figure 17.12, Fundamentals of Database Systems, Elmasri and Navathe, 7th Edition, 2016

Searching for 7 in the B^+-Tree



We only keep the root node of the B+-tree in the main memory.



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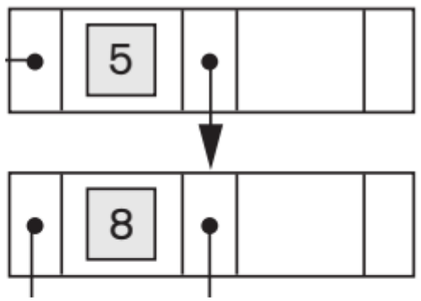
Searching for 7 in the B$^+$-Tree (contd.)



Since 7 > 5, we bring the node pointed to by 5's right tree pointer from the disk.

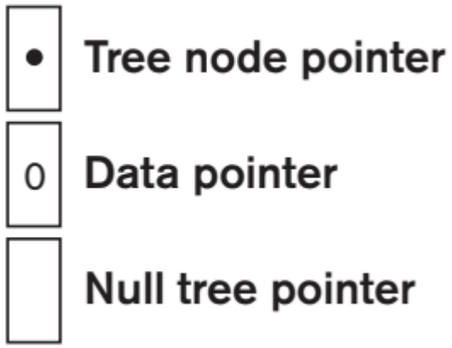
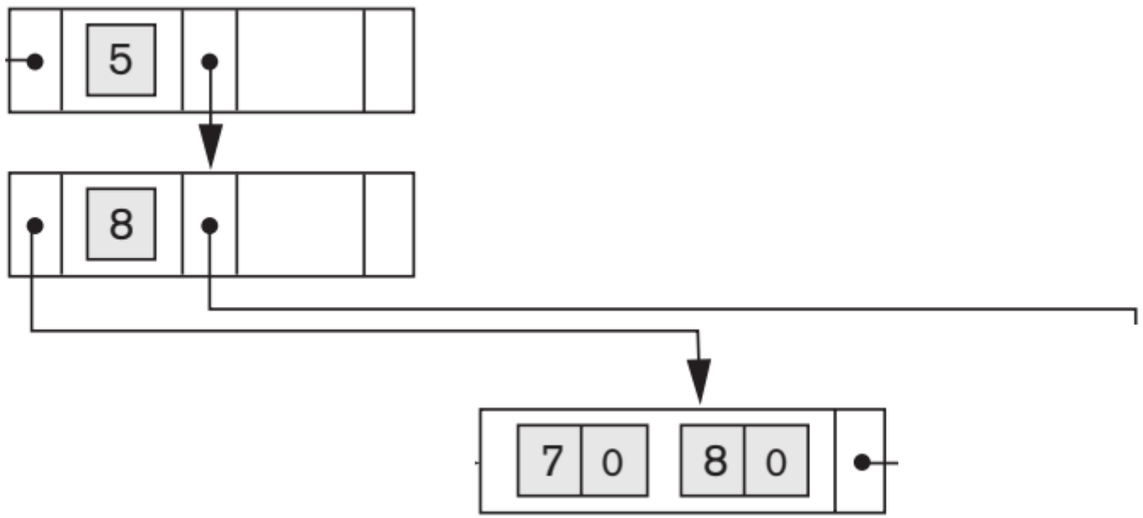
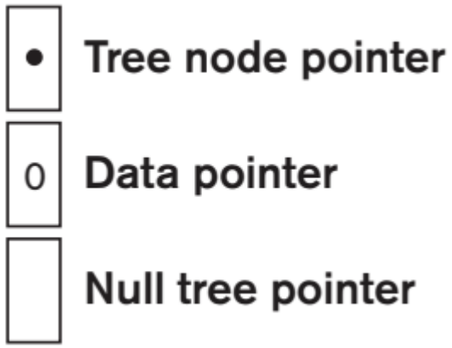


Figure 17.12, Fundamentals of Database Systems, Elmasri and Navathe, 7th Edition, 2016

Searching for 7 in the B$^+$-Tree (contd.)





Since 7 < 8, we bring the node pointed to by 8's left tree pointer from the disk. Thus, we get access to index 7 and its satellite data pointer in **only 2 disk accesses**. Such a low number of disk accesses is made possible by **keeping the search tree height-balanced** and **simultaneously maximising the number of keys in each node**.

Figure 17.12, Fundamentals of Database Systems, Elmasri and Navathe, 7th Edition, 2016

Summary

Today we discussed two height-balanced **m-ary** search trees, namely, the B-tree and the B^+-tree.

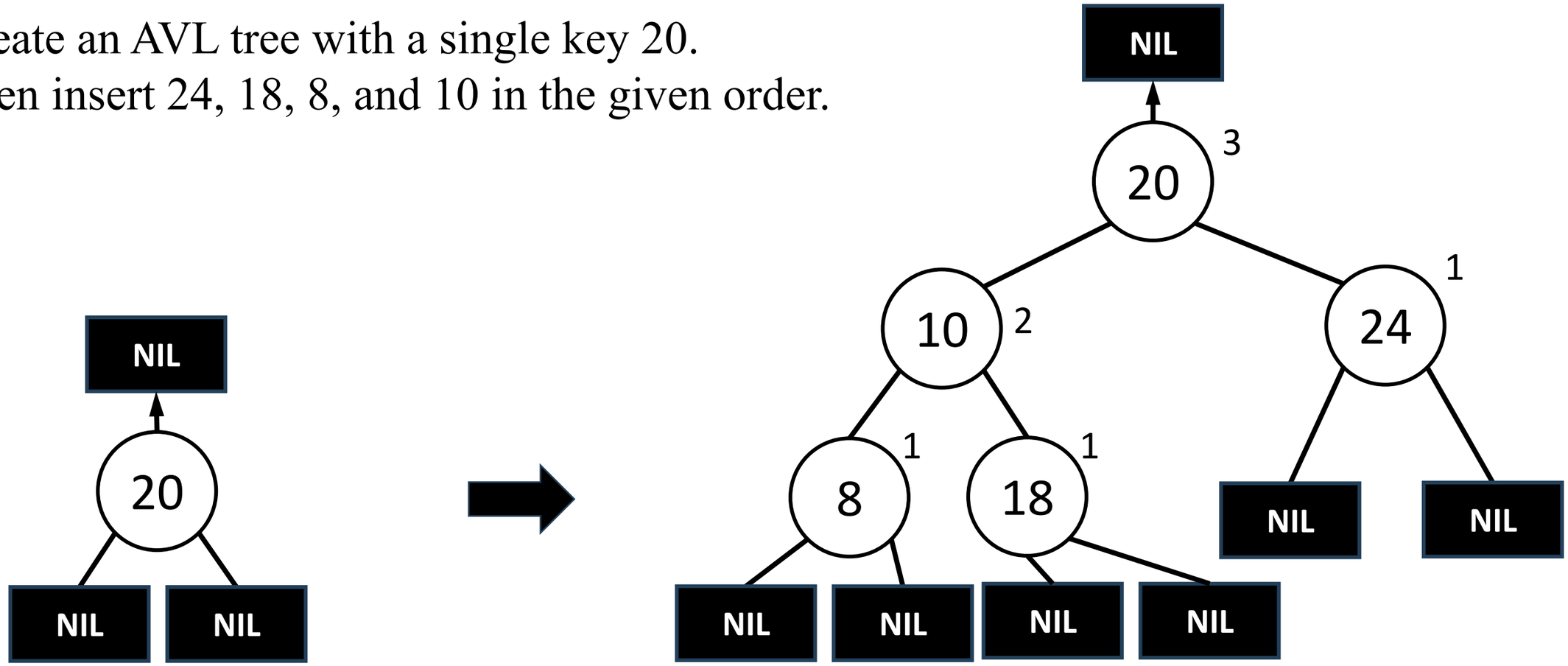
Next Class

We will discuss another very useful search tree known as **Trie** (pronounced ‘tree’ or ‘try’ by different people).

We will also discuss some graph algorithms for finding the **minimum spanning tree (MST)** of a graph and the **shortest paths** between a pair of nodes.

Today's Practice Problem

Create an AVL tree with a single key 20.
Then insert 24, 18, 8, and 10 in the given order.



Thank You